

Stat 240 Midterm

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```
library(tidyverse)
library(RSQLite)
library(DBI)
```

```
download.file("https://owenward.github.io/files/nhl_data.tsv", "nhl_data.tsv")
```

Problem 1

(a)

```
nhl_data <- read.table("nhl_data.tsv", header=TRUE, sep = "\t")

nhl_sub <- nhl_data |>
  filter(W > 50)

head(nhl_data)
```

	Team	GP	SOG	W	L	OTL	PTS	Reg_PTS	PPG	SO_PTS	Pct_SO	RW	ROW	SOW	SOL
1	LA	82	9	47	25	10	104	98	1.342466	6	0.6666667	37	41	6	3
2	STL	82	6	37	38	7	81	78	1.026316	3	0.5000000	27	34	3	3
3	EDM	82	4	50	23	9	109	109	1.397436	0	0.0000000	45	50	0	4
4	CGY	82	7	38	27	17	93	91	1.213333	2	0.2857143	31	36	2	5
5	PIT	82	2	40	31	11	91	90	1.125000	1	0.5000000	31	39	1	1
6	DET	82	7	35	37	10	80	77	1.026667	3	0.4285714	28	32	3	4
	GF	GA	DIFF				CONF								
1	280	257	23				Western								
2	263	301	-38				Eastern								

```
3 325 260    65 Western
4 260 252     8 Western
5 262 264    -2 Eastern
6 240 279   -39 Western
```

```
head(nhl_sub)
```

```
      Team GP SOG  W  L OTL PTS Reg_PTS      PPG SO_PTS      Pct_SO RW ROW SOW SOL
1  CAR 82   7 52 21   9 113      109 1.453333      4 0.5714286 39 48  4  3
2  COL 82   9 51 24   7 109      103 1.410959      6 0.6666667 36 45  6  3
      GF  GA DIFF      CONF
1 266 213   53 Eastern
2 280 226   54 Western
```

```
dim(nhl_sub)[1]
```

```
[1] 2
```

(b)

```
nhl_mod <- nhl_sub |>
  select(-c("GF", "GA", "DIFF"))
head(nhl_mod)
```

```
      Team GP SOG  W  L OTL PTS Reg_PTS      PPG SO_PTS      Pct_SO RW ROW SOW SOL
1  CAR 82   7 52 21   9 113      109 1.453333      4 0.5714286 39 48  4  3
2  COL 82   9 51 24   7 109      103 1.410959      6 0.6666667 36 45  6  3
      CONF
1 Eastern
2 Western
```

(c)

```
write.table(nhl_mod[, -1], file = "nhl.psv", sep = "._.")
```

Problem 2

(a)

```
db <- dbConnect(SQLite(), dbname = "nhl.sqlite")
dbWriteTable(db, name = "NHL", value = nhl_data, overwrite = TRUE)
dbListTables(db)
```

```
[1] "NHL"
```

(b)

```
query <- "SELECT * FROM NHL WHERE W > 50"
sql_nhl_data <- dbGetQuery(db, query)
sql_nhl_data
```

	Team	GP	SOG	W	L	OTL	PTS	Reg_PTS	PPG	SO_PTS	Pct_SO	RW	ROW	SOW	SOL
1	CAR	82	7	52	21	9	113	109	1.453333	4	0.5714286	39	48	4	3
2	COL	82	9	51	24	7	109	103	1.410959	6	0.6666667	36	45	6	3
	GF	GA	DIFF	CONF											
1	266	213	53	Eastern											
2	280	226	54	Western											

```
# show that the number of rows is equal with nhl_mod from Q1
(dim(sql_nhl_data)[1] == dim(nhl_mod)[1])
```

```
[1] TRUE
```

(c)

```
query2 <- "SELECT CONF, AVG(W) AS avg_wins, COUNT(Team) AS num_teams
          FROM NHL
          GROUP BY CONF"
dbGetQuery(db, query2)
```

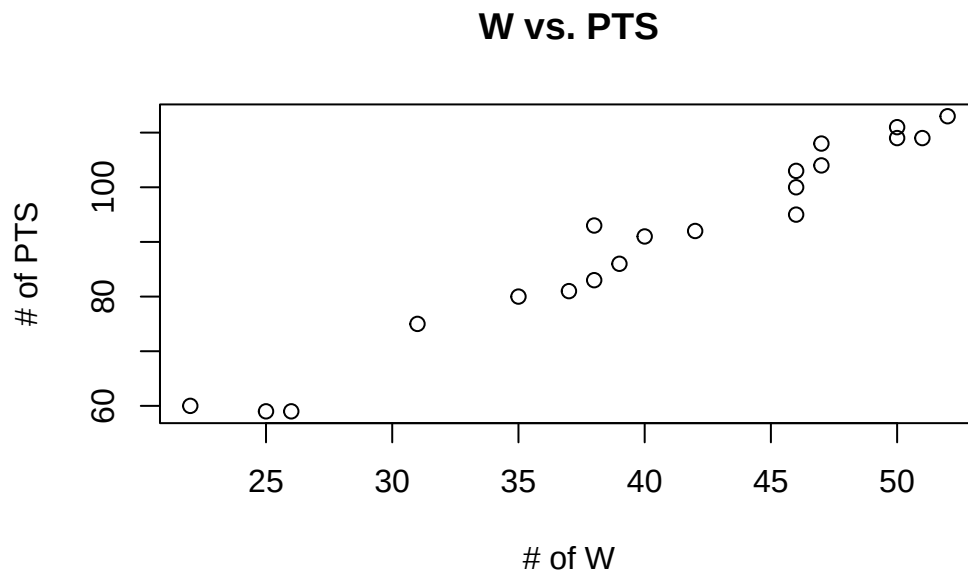
	CONF	avg_wins	num_teams
1	Eastern	37.55556	9
2	Western	42.72727	11

Problem 3

(a)

```
num_W <- nhl_data$W
num_PTS <- nhl_data$PTS

plot(num_W, num_PTS, xlab = "# of W", ylab = "# of PTS", main = "W vs. PTS")
```



(b)

```
fit <- lm(W ~ PTS, data = nhl_data)
alpha <- coef(fit)[1]
beta <- coef(fit)[2]

alpha
```

(Intercept)
-5.438311

```
beta
```

```
PTS  
0.506221
```

(c)

```
SSE <- function(pars = c(0, 0), x, y) {  
  alpha <- pars[1]  
  beta <- pars[2]  
  f_x <- alpha + beta * x  
  SSE <- sum((y - f_x)^2)  
  SSE  
}  
  
# optimized SSE  
SSE(pars = c(alpha, beta), x = num_W, y = num_PTS)
```

```
[1] 117342.3
```

for testing non-optimal SSEs:

```
# change the alpha to non-optimal:  
SSE(pars = c(-10, beta), x = num_W, y = num_PTS)
```

```
[1] 131541.6
```

```
# change the beta to non-optimal:  
SSE(pars = c(alpha, 10), x = num_W, y = num_PTS)
```

```
[1] 1998519
```

```
# change both to non-optimal:  
SSE(pars = c(5, 10), x = num_W, y = num_PTS)
```

```
[1] 2129303
```